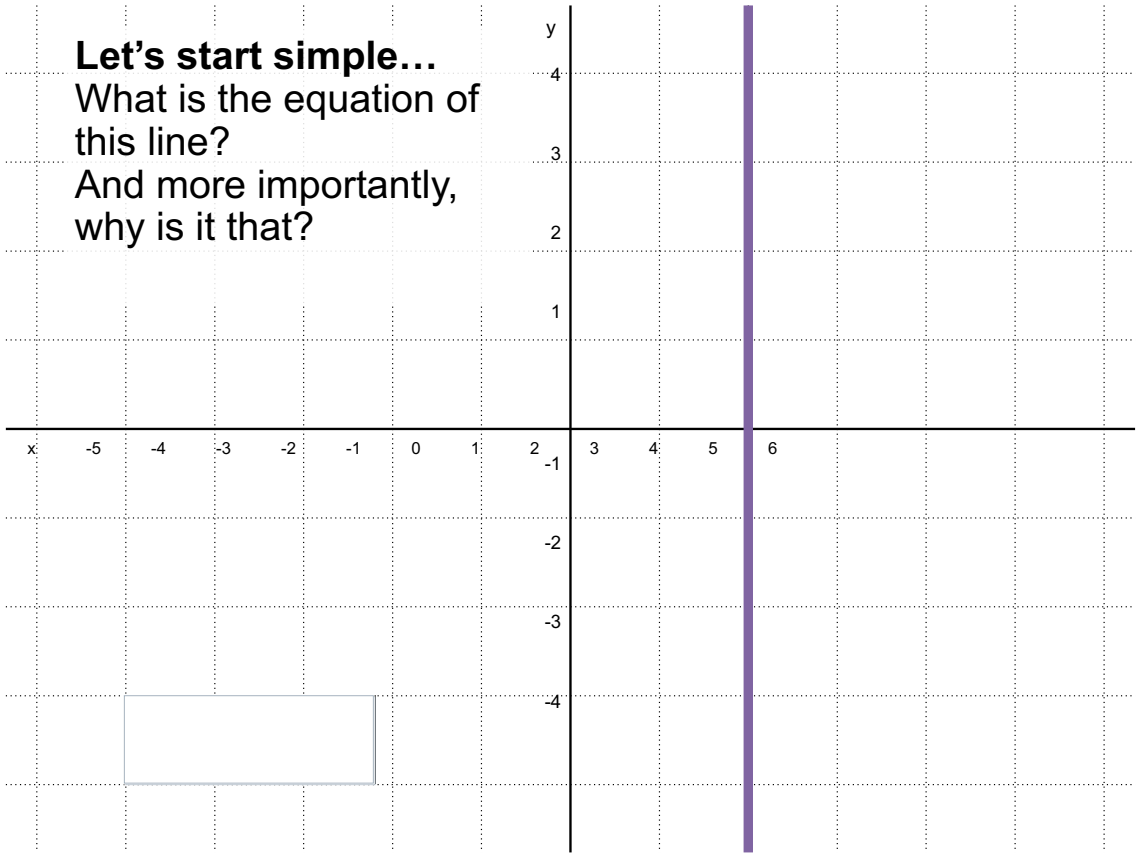
## Coordinate Geometry

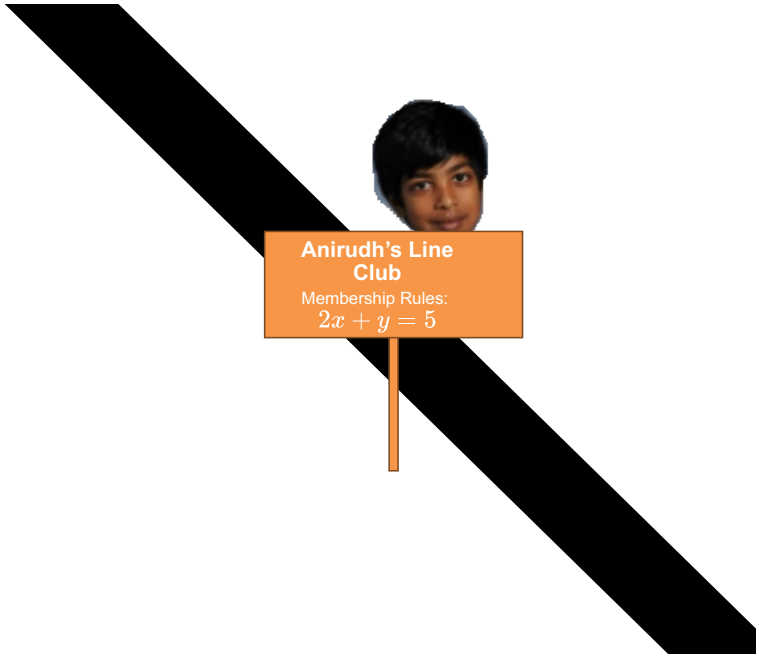
Chapter 5 · Original slide 4




Chapter 5 · Original slide 5

# Lines and Equations of Lines

! A line consists of all points which satisfy some equation in terms of $x$ and/or $y$.



## Coordinate Geometry

Chapter 5 · Original slide 6

# Examples

This means we can **substitute** the values of a coordinate into our equation whenever we know the point lies on the line.

The point $(5, a)$ lies on the line with equation $y = 3x + 2$. Determine the value of $a$.

Find the coordinate of the point where the line $2x + y = 5$ cuts the $x$-axis.

Chapter 5 · Original slide 7

# Test Your Understanding

Determine where the line $x + 2y = 3$ crosses the:

a) $y$-axis:

b) $x$-axis:

What mistakes do you think it's easy to make?

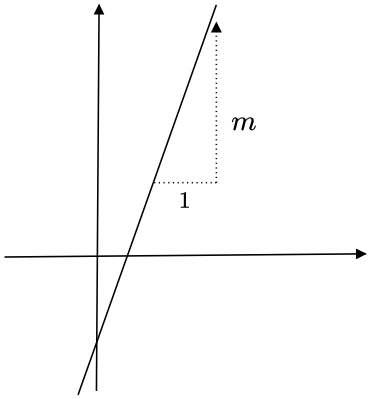
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- **S**

## Coordinate Geometry

Chapter 5 · Original slide 8

### Recap of gradient

The steepness of a line is known as the **gradient**.
It tells us what $y$ changes by as $x$ increases by 1.




So if the $y$ value increased by 6 as the $x$ value increased by 2, what is $y$ increasing by for each unit increase of $x$?
How would that give us a suitable formula for the gradient $m$?

$m =$ ☐

Chapter 5 · Original slide 9

### Examples

Find the gradient of the line that goes through the points:

1 $(1, 4)$ $(3, 10)$ $m =$ ☐

2 $(5, 7)$ $(8, 1)$ $m =$ ☐

3 $(2, 2)$ $(-1, 10)$ $m =$ ☐

4 Show that the points
$\mathrm{A}\,(3, 4), B\,(5, 5), C\,(11, 8)$ all lie on a straight

## Coordinate Geometry

Chapter 5 · Original slide 10

### Further Example

The line joining $(2, -5)$ to $(4, a)$ has gradient -1. Work out the value of $a$.

$\frac{a}{}$

$\frac{a}{}$

$a$

Chapter 5 · Original slide 11

### $y = mx + c$

One form we can put a straight line equation in is:

$$y = mx + c$$

Gradient

$y$-intercept

Why does it work?

## Coordinate Geometry

Chapter 5 · Original slide 12

# $y = mx + c$

Determine the gradient and $y$-intercept of the line with equation $4x - 3y + 5 = 0$

$3y$

$y =$

Make $y$ the subject so we have the form $y = mx + c$
Put $y$ on the side it's positive.

Chapter 5 · Original slide 13

# $ax + by + c = 0$

At GCSE, $\mathbf{y=mx+c}$ was the main form you would express a straight line equation, sometimes known as the '**slope-intercept form**'.
But another common form is $\mathbf{ax+by+c=0}$, where $a, b, c$ are integers.
This is known as the '**standard**' form.

Express $y = \frac{1}{3}x - \frac{2}{3}$ in the form $ax + by + c = 0$, where $a, b, c$ are integers.

We'll see on the next slide WHY we might want to put an equation in this form over $y = mx + c$…

## Coordinate Geometry

Chapter 5 · Original slide 14

# Just for your interest…

Why might we want to put a straight line equation in the form $ax + by + c = 0$?

$y = mx + c$
"Slope-Intercept Form"

$ax + by + c = 0$
"Standard Form"

### Coverage

$y = mx + c$ doesn't allow you to represent vertical lines. Standard form allows us to do this by just making $b$ zero.

$x + 4 = 0$

### Symmetry

In general, the '**linear combination**' of two variables $x$ and $y$ is $ax + by$, i.e. "some amount of $x$ and some amount of $y$". There is a greater elegance and symmetry to this form over $y = mx + c$ because $x$ and $y$ appear similarly within the expression.

### Usefulness

This more 'elegant' form also means it ties in with vectors and matrices. In FM, you will learn about the '**dot product**' of two vectors:

$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = ax + by$$

thus since $ax + by + c = 0$, we can represent a straight line using:

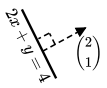
$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} + c = 0 \qquad (1)$$

We can extend to 3D points to get the equation of a **plane**:

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} + d = 0 \qquad (2)$$

Conveniently, in equation $(1)$, the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ is **perpendicular to the line**. And in equation $(2)$, the vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is perpendicular to the plane. Nice!

$2x + y = 4$ ➡ $\begin{pmatrix} 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = 4$ ➡




Chapter 5 · Original slide 15

# Test Your Understanding

Express $y = \frac{2}{5}x + \frac{3}{5}$ in the form $ax + by + c = 0$, where $a, b, c$ are integers.

$5y$
$2x$

# Coordinate Geometry

Chapter 5 · Original slide 16

## Exercise 5A – Pure Y1 Page 91

9 Show that the points $A(2, 3)$, $B(4, 4)$ and $C(10, 7)$ can be joined by a straight line.

10 Show that the points $A(-2a, 5a)$, $B(0, 4a)$ and points $C(6a, a)$ are collinear. $(3\ \mathrm{ma}\ldots$

Chapter 5 · Original slide 17

## Exercise 5B – Pure Y1 Page 92-93

8 A line $l$ cuts the $x$-axis at $(5, 0)$ and the $y$-axis at $(0, 2)$.

a Find the gradient of the line.

b Find an equation of the line in the form $ax + by + c = 0$.

9 Show that the line with equation $ax + by + c = 0$ has gradient $-\frac{a}{b}$ and cuts the $y$-axis at $-\frac{c}{b}$.

10 The line $l$ with gradient 3 and $y$-intercept $(0, 5)$ has the equation $ax - 2y + c = 0$.

Find the values of $a$ and $c$. (2 marks)

## Coordinate Geometry

Chapter 5 · Original slide 18

# Exercise 5B – Pure Y1 Page 92-93

11 The straight line $l$ passes through $(0, 6)$ and has gradient $-2$. It intersects the line with equation $5x - 8y - 15 = 0$ at point $P$. Find the coordinates of $P$.

12 The straight line $l_1$ with equation $y = 3x - 7$ intersects the straight line $l_2$ with equation $ax + 4y - 17 = 0$ at the point $P(-3, b)$.

a Find the value of $b$.

b Find the value of $a$.

Chapter 5 · Original slide 19

# Equations using two points/point + gradient

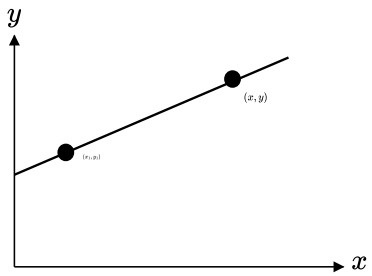
Find the equation of the line that goes through $(3, 5)$ and has gradient 2.

How would you have done this at **GCSE**?

## Coordinate Geometry

Chapter 5 · Original slide 20

# A new way…




**Notes**: Note that $x_1$ and $y_1$ are constants while $x$ and $y$ are variables. The latter are variables because as these 'vary', we get different points on the line.

Suppose that $(x_1, y_1)$ is some fixed point on the line that we specify (e.g. $(3, 5)$).
Suppose that $(x, y)$ represents a generic point on the line, which is allowed to change as we consider different points on this line.
Then:

$m = \frac{y-y_1}{x-x_1}$

Thus:

Chapter 5 · Original slide 21

# Quickfire Questions

In a nutshell: You can use this formula whenever you have (a) a gradient and (b) any point on the line.

| Gradient | Point | (Unsimplified) Equation |
|---|---|---|
| $3$ | $(1, 2)$ | |
| $5$ | $(3, 0)$ | |
| $2$ | $(-3, 4)$ | |
| $\frac{1}{2}$ | $(1, -5)$ | |
| $9$ | $(-4, -4)$ | |

## Coordinate Geometry

Chapter 5 · Original slide 22

# Using 2 points

Find the equation of the line that goes through $(4, 5)$ and $(6, 2)$, giving your equation in the form $ax + by + c = 0$.

**Test Your Understanding:**
Find the equation of the line that goes through $(-1, 9)$ and $(4, 5)$, giving your equation in the form $ax + by + c = 0$.

Chapter 5 · Original slide 23

# Exercise 5C – Pure Y1 Page 94-95

3 Find the equation of the line $l$ which passes through the points $A(7, 2)$ and $B(9, -8)$. Give your answer in the form $ax + by + c = 0$.

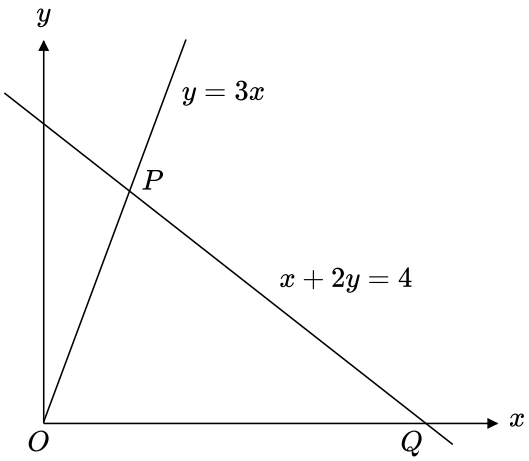
5 T $l$ $(a, 4)$ $(3a, 3)$. An equation of $l$ is $x + 6y + c = 0$. Find the value of $a$ and the value of $c$.

6 The straight line $l$ passes through $(7a, 5)$ and $(3a, 3)$. An equation of $l$ is $x + by - 12 = 0$. Find the value of $a$ and the value of $b$.

# Coordinate Geometry

Chapter 5 · Original slide 24

## Intersection of lines




The diagram shows two lines with equations $y = 3x$ and $x + 2y = 4$, which intersect at the point $P$.

a) Determine the coordinates of $P$.

**We know from Chapter 4 we**

b

b

Chapter 5 · Original slide 25

## Test Your Understanding

C1 Edexcel May 2013 Q6

The straight line $L_1$ passes through the points $(-1, 3)$ and $(11, 12)$.
(a) Find an equation for $L_1$ in the form $ax + by + c = 0$, where $a, b$ and $c$ are integers.
\hfill(4)
The line $L_2$ has equation $3y + 4x - 30 = 0$.
(b) Find the coordinates of the point of intersection of $L_1$ and $L_2$. \hfill(3)

$(-1, 3)$,
**(a)**
$m = \frac{y_2}{x_2}$
M1: Corr
$y - 3 =$
Correct s
$4y - 3x$
Or equiv
This A1 s

**(b)** Solve
Must rea
M1
$x = 3$ or
\textbf{B
**Fully cor**

## Coordinate Geometry

Chapter 5 · Original slide 26

# Exercise 5D – Pure Y1 Page 96

6 The line that passes through the points $(-3, -5)$ and $(4, 9)$ meets the $y$-axis at the point $G$. Work out the coordinates of the point $G$.

9 The lines $y = 4x - 10$ and $y = x - 1$ intersect at the point $T$. Find the equation of the line with gradient $-\frac{2}{3}$ that passes through the point $T$. Write your answer in the form $ax + by + c = 0$, where $a, b$ and $c$ are integers.

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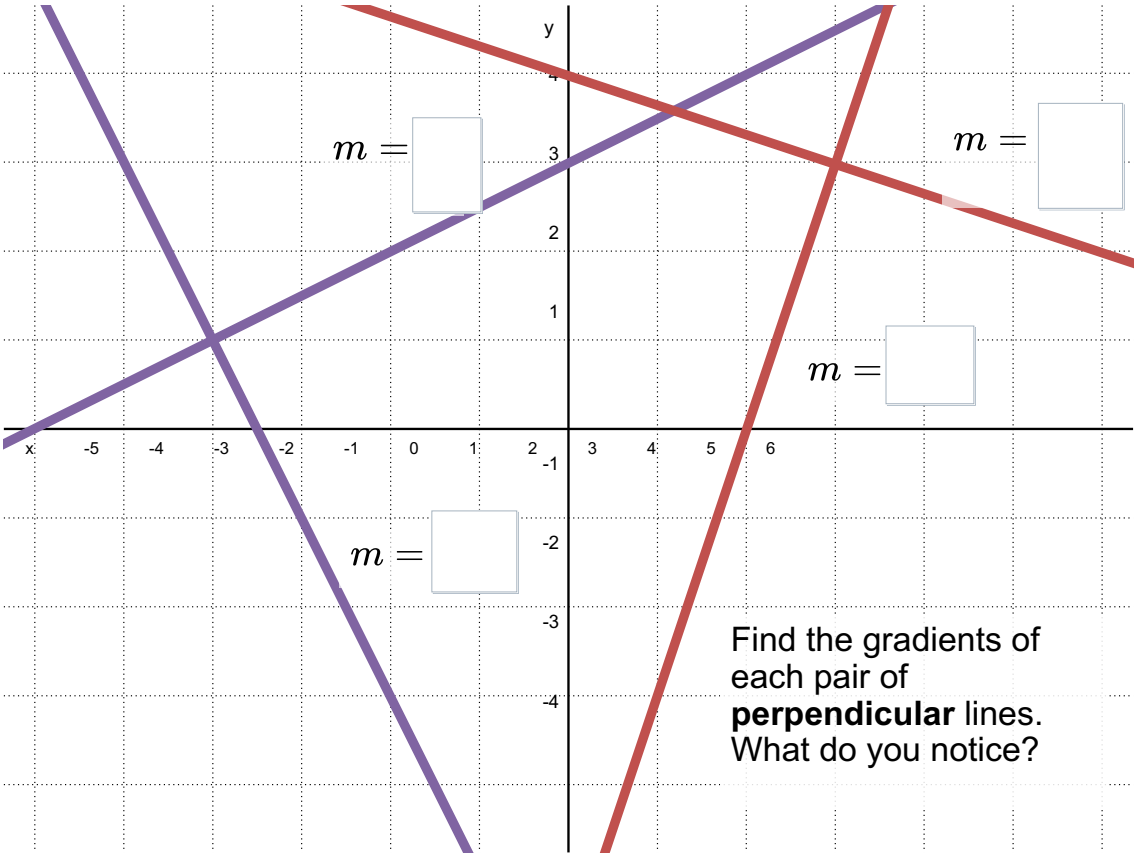
# Exercise 5D – Pure Y1 Page 96

14 The line $y = 4x + 5$ meets the $y$-axis at the point $C$. The line $y = -3x - 15$ meets the $x$-axis at the point $D$. Find the equation of the line joining the points $C$ and $D$. Write your answer in the form $ax + by + c = 0$, where $a, b$ and $c$ are integers.

16 The lines $y = -2x + 1$ and $y = x + 7$ intersect at the point $L$. The point $M$ has coordinates $(-3, 1)$. Find the equation of the line that passes through the points $L$ and $M$.

## Coordinate Geometry

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Chapter 5 · Original slide 29

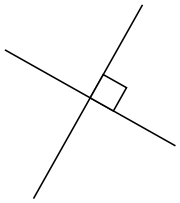
# Perpendicular Lines

The gradients of parallel lines are equal.
If two lines are perpendicular, then the gradient of one is the **negative reciprocal** of the other.

$$m_1 = -\frac{1}{m_2}$$

To **show** that two lines are perpendicular:

$$m_1 m_2 = -1$$



| Gradient | |
|---|---|
| $2$ | |
| $-3$ | |
| $\frac{1}{4}$ | |
| $5$ | |
| $-\frac{2}{7}$ | |
| $\frac{7}{5}$ | |

## Coordinate Geometry

Chapter 5 · Original slide 30

### Example Problems

1 A line is goes through the point (9,10) and is perpendicular to another line with equation $y = 3x + 2$. What is the equation of the line?

2 A line $L_1$ goes through the [illegible] $A(1, 3)$ $B(3, -1)$. A second line $L_2$ is perpendicular to $L_1$ and passes through point B. Where does $L_2$ cross the x-axis?

$(\mathbf{5,0})$

3 Are the following lines para

$y = \frac{1}{2}x$

$2x - y + 4 = 0$

**Neither. Gradients are $\frac{1}{2}$ and 2. But $\frac{1}{2} \times 2 = 1$, not -1, so not perpendicular.**

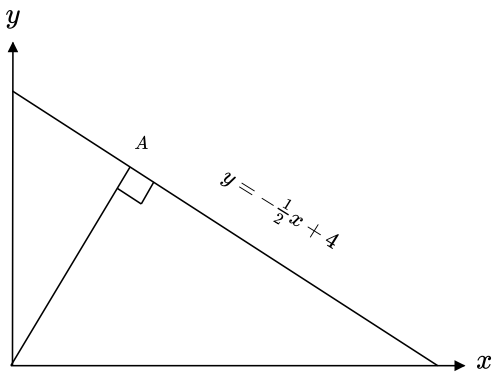
Chapter 5 · Original slide 31

### Test Your Understanding

1 A line goes through the point (4,7) and is perpendicular to another line with equation $y = 2x + 2$. What is the equation of the line? Put your answer in the form $ax + by + c = 0$, where $a, b, c$ are integers.

$\mathbf{y = -\frac{1}{2}x + 9}$

2 Determine the point $A$.



# Coordinate Geometry

Chapter 5 · Original slide 32

## Exercise 5E/5F – Pure Y1 Page 96

3 The coordinates of a quadrilateral $ABCD$ are $A(-6, 2), B(4, 8), C(6, 1)$ and $D(-9, -8)$. Show that the quadrilateral is a trapezium.

7 A line is parallel to the line $2x - 3y - 1 = 0$ and it passes through the point $(0, 0)$. Write down the equation of the line.

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## Exercise 5F – Pure Y1 Page 100

9 The line $l$ passes through the points $(-3, 0)$ and $(3, -2)$ and the line $n$ passes through the points $(1, 8)$ and $(-1, 2)$. Show that the lines $l$ and $n$ are perpendicular.

10 The vertices of a quadrilateral $ABCD$ have coordinates $A(-1, 5), B(7, 1), C(5, -3)$ and $D(-3, 1)$. Show that the quadrilateral is a rectangle.

## Coordinate Geometry

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# Exercise 5E/5F – Pure Y1 Page 96

**Extension Problems**

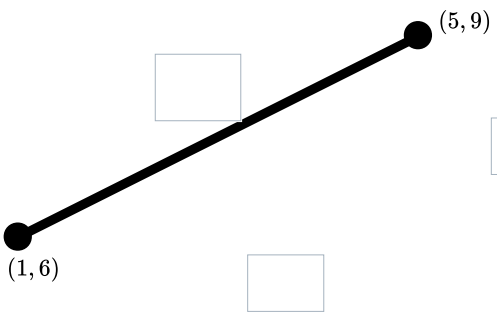
*[MAT 2004 1D]*
What is the reflection of the point $(3, 4)$ in the line $3x + 4y = 50$?

*[MAT 2014 1D]* The reflection of the point $(1, 0)$ in the line $y = mx$ has coordinates: (in terms of $m$)

*[STEP I 2004 Q6]* The three points $A, B, C$ have coordinates $(p_1, q_1), (p_2, q_2)$ and $(p_3, q_3)$, respectively. Find the point of intersection of the line joining $A$ to the midpoint of $BC$, and the line joining $B$ to the midpoint of $AC$. Verify that this point lies on the line joining $C$ to the midpoint of $AB$.
The point $H$ has coordinates $(p_1 + p_2 + p_3, q_1 + q_2 + q_3)$. Show that if the line $AH$ intersects the line $BC$ at right angles, then $p_2^2 + q_2^2 = p_3^2 + q_3^2$, and write down a similar result if the line $BH$ intersects the line $AC$ at right angles.
Deduce that if $AH$ is perpendicular to $BC$ and also $BH$ is perpendicular to $AC$, then $CH$ is perpendicular to $AB$.

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# Distances between points




How could we find the **distance** between these two points?
Hint:

## Coordinate Geometry

Chapter 5 · Original slide 36

# Examples

**Distance between:**

$(3, 4)$ and $(5, 7)$
$(5, 1)$ and $(6, -3)$
$(0, -2)$ and $(-1, 3)$

**Note**: Unlike with gradient, we don't care if the difference is positive or negative (it's being squared to make it positive anyway!)

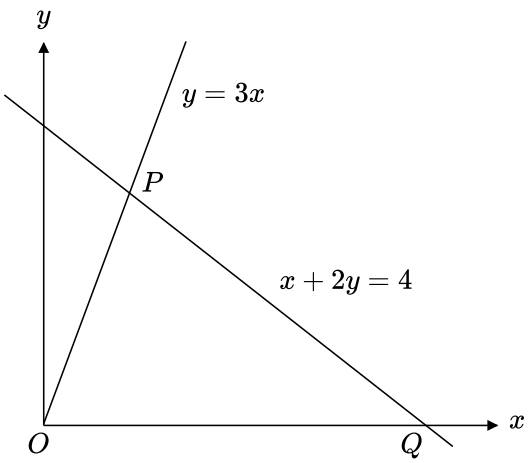
Quickfire Questions:

**Distance between:**

$(1, 10)$ and $(4, 14)$
$(3, -1)$ and $(0, 1)$
$(-4, -2)$ and $(-12, 4)$

Chapter 5 · Original slide 37

# Area of Shapes




The diagram shows two lines with equations $y = 3x$ and $x + 2y = 4$, which intersect at the point $P$.

a) Determine the coordinates of $P$.

*(We did this in a previous lesson)*
**Just solve two equations simultaneously.**

$$\mathbf{x+2\,(3x)=4}$$
$$\mathbf{7x=4}$$
$$\mathbf{x=\frac{4}{7}}$$
$$\mathbf{y=3\,\left(\frac{4}{7}\right)=\frac{12}{7}}$$

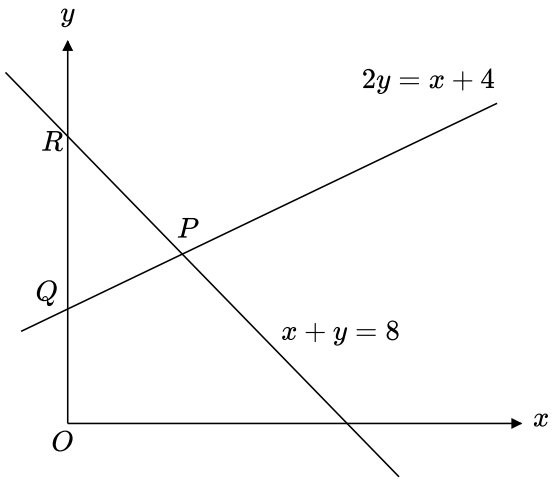
b) The line $x + 2y = 4$ intersects the $x$-axis at the point $Q$. Determine the area of the triangle $OPQ$.

**When $\mathbf{y=0, x=4}$**

$$\text{Area} = \frac{1}{2} \times 4 \times \frac{12}{7} = \frac{24}{7}$$

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## Further Example



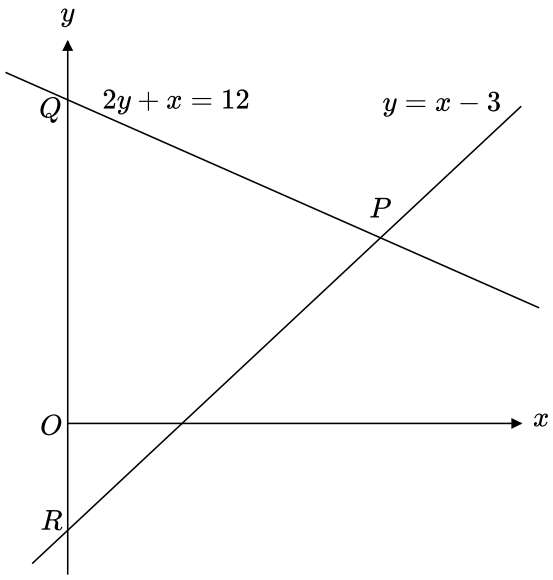

**Tip**: When finding areas of triangles in exam questions, one line is often vertical or horizontal. You should generally choose this to be the 'base' of your triangle.

a Determine the length of $PQ$.

b Determine the area $PQR$.

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## Test Your Understanding




a) Determine the coordinate of $P$.

$\mathbf{2}$ (illegible)

$\mathbf{x}=$

b) Determine the area of $PQR$.

$\mathbf{R}\,(0,-3)$

$\mathbf{Q}$

$\mathbf{A}$

c)

$\mathbf{P}\,(6,3), \quad \mathbf{Q}\,(0,6)$

$\mathbf{Dist} = \sqrt{6^2+3^2} = \sqrt{45}$

## Coordinate Geometry

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# Exercise 5G – Pure Y1 Page 102

7 A point $P$ lies on the line with equation $y = 4 - 3x$. The point $P$ is a distance $\sqrt{34}$ from the origin. Find the two possible positions of point $P$. \hfill(5 ma...

8 The vertices of a triangle are $A(2, 7)$, $B(5, -6)$ and $C(8, -6)$.
a Show that the triangle is a scalene triangle.
b Find the area of the triangle $ABC$.

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# Exercise 5G – Pure Y1 Page 103

12 The straight line $l_1$ passes through the point $(-4, 14)$ and has gradient $-\frac{1}{4}$.
a Find an equation for $l_1$ in the form $ax + by + c = 0$, where $a, b$ and $c$ are integers.
b Write down the coordinates of $A$, the point where straight line $l_1$ crosses the $y$-axis.
The straight line $l_2$ passes through the origin and has gradient 3. The lines $l_1$ and $l_2$ intersect at the point $B$.
c Calculate the coordinates of $B$.
d Calculate the exact area of $\triangle OAB$.

## Coordinate Geometry

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# Exercise 5G – Pure Y1 Page 102-103

**Extension Problems**

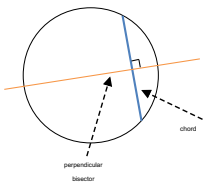
*[MAT 2001 1C]*
The shortest distance from the origin to the line $3x + 4y = 25$ is what?

Chapter 6 · Original slide 4

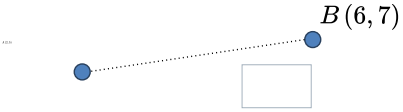
# Midpoints and Perpendicular Bisectors

4

Later in the chapter you will need to find the perpendicular bisector of a chord of a circle.




**What two properties does a perpendicular bisector of two points $\mathbf{A}$ and $\mathbf{B}$ have?**



# Coordinate Geometry

Chapter 6 · Original slide 5

## Test Your Understanding

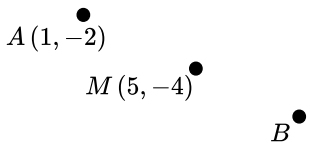
5

Find the perpendicular bisector of the line $AB$ where $A$ and $B$ have the coordinates:
a) $A\,(4,7), B\,(10,17)$
b) $A\,(x_1,y_1), B\,(x_2,y_2)$

$M\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$

$m_{AB}=\frac{y_2-y_1}{x_2-x_1}$ $\quad m_{\perp}=\frac{x_1-x_2}{y_2-y_1}$

A line segment $AB$ is the diameter of a circle with centre $(5,-4)$. If $A$ has coordinates $(1,-2)$, what are the coordinates of $B$?




Chapter 6 · Original slide 6

## Exercise 6A - Y1 Page 115

6

6 The line segment $JK$ is a diameter of a circle, where $J$ and $K$ are $(\frac{3}{4},\frac{4}{3})$ and $(-\frac{1}{2},2)$ respectively. The centre of the circle lies on the line segment with equation $y=8x+b$. Find the value of $b$.

7 The line segment $AB$ is a diameter of a circle, where $A$ and $B$ are $(0,-2)$ and $(6,-5)$ respectively. Show that the centre of the circle lies on the line $x-2y-10=0$.

# Coordinate Geometry

Chapter 6 · Original slide 7

## Exercise 6B - Y1 Page 116

7

2 The line $FG$ is a diameter of the circle centre $C$, where $F$ and $G$ are $(-2, 5)$ and $(2, 9)$ respectively. The line $l$ passes through $C$ and is perpendicular to $FG$. Find the equation of $l$. \hfill(7 marks)

3 The line $JK$ is a diameter of the circle centre $P$, where $J$ and $K$ are $(0, -3)$ and $(4, -5)$ respectively. The line $l$ passes through $P$ and is perpendicular to $JK$. Find the equation of $l$. Write your answer in the form $ax + by + c = 0$, where $a, b$ and $c$ are integers.

Chapter 6 · Original slide 8

## Exercise 6B - Y1 Page 117

8

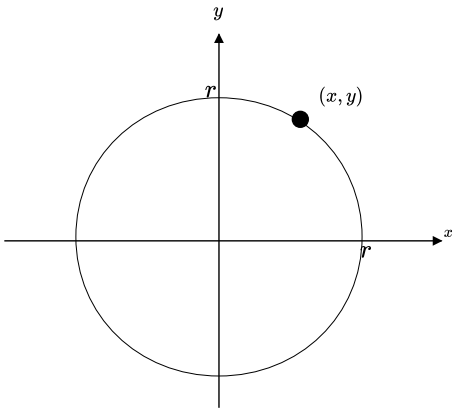
5 Point $X$ has coordinates $(7, -2)$ and point $Y$ has coordinates $(4, q)$. The perpendicular bisector of $XY$ has equation $y = 4x + b$. Find the value of $q$ and the value of $b$.

# Coordinate Geometry

Chapter 6 · Original slide 9

## Equation of a circle

9



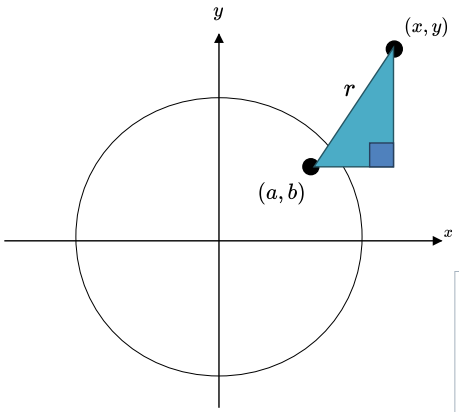

Recall that a line can be a set of points $(x, y)$ that satisfy some equation. Suppose we have a point $(x, y)$ on a circle centred at the origin, with radius $r$. What equation must $(x, y)$ satisfy?

(Hint: draw a right-angled triangle inside your circle, with one vertex at the origin and another at the circumference)

Chapter 6 · Original slide 10

## Equation of a circle

10




Now suppose we shift the circle so it's now centred at $(a, b)$. What's the equation now?

(Hint: What would the sides of this right-angled triangle be now?)

## Coordinate Geometry

Chapter 6 · Original slide 11

### Quickfire Questions

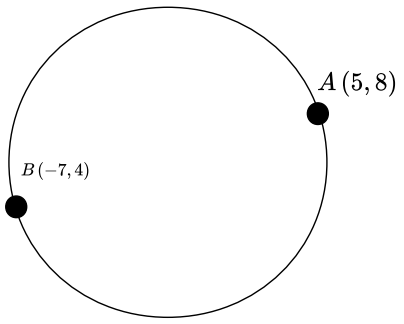
11

| Centre | Radius | Equation |
|---|---|---|
| $(0, 0)$ | $5$ | |
| $(1, 2)$ | $6$ | |
| | | $(x+3)^2+(y-5)^2=1$ |
| | | $(x+5)^2+(y-2)^2=49$ |
| | | $(x+6)^2+y^2=16$ |
| | | $(x-1)^2+(y+1)^2=3$ |
| | | $(x+2)^2+(y-3)^2=8$ |

Chapter 6 · Original slide 12

### Finding the equation using points

12




A line segment $AB$ is the diameter of a circle, where $A$ and $B$ have coordinates $(5, 8)$ and $(-7, 4)$ respectively. Determine the equation of the circle.

Hint: What two things do we need to use the circle formula?

## Coordinate Geometry

Chapter 6 · Original slide 13

### Test Your Understanding

13

Edexcel C2 Jan 2005 Q2

The points $A$ and $B$ have coordinates $(5, -1)$ and $(13, 11)$ respectively.
(a) Find the coordinates of the mid-point of $AB$. \hfill(2)
Given that $AB$ is a diameter of the circle $C$,
(b) find an equation for $C$. \hfill(4)

a)

b)

Chapter 6 · Original slide 14

### Completing the square

14

When the equation of a circle is in the form $(x-a)^2 + (y-b)^2 = r^2$, we can instantly read off the centre $(a, b)$ and the radius $r$.
But what if the equation wasn't in this form?

Find the centre and radius of the circle with equation $x^2 + y^2 - 6x + 2y - 6 = 0$

**Hint**: Have we seen a method in a previous chapter that allows us to turn a $x^2$ term and a $x$ term into a single expression involving $x$?

# Coordinate Geometry

Chapter 6 · Original slide 15

## Further Example

15

Edexcel C2 June 2012 Q3a,b

The circle $C$ with centre $T$ and radius $r$ has equation
$x^2 + y^2 - 20x - 16y + 139 = 0$

(a) Find the coordinates of the centre of $C$. \hfill(3)
(b) Show that $r = 5$. \hfill(2)

Chapter 6 · Original slide 16

## Example

16

Given the generic equation of a circle $x^2 + y^2 + ax + by + c = 0$ find a formula to determine the co-ordinate of the centre and the radius of the circle.

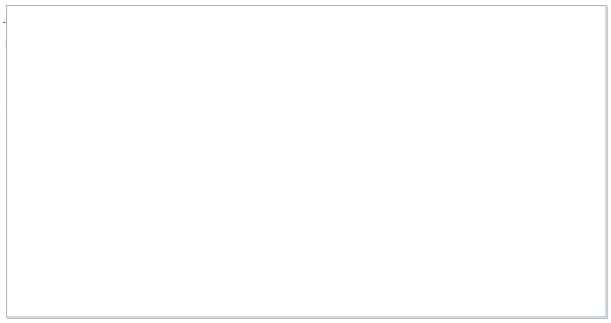
# Coordinate Geometry

Chapter 6 · Original slide 17

## Exercise 6C - Y1 Page 120

17

5 The line $PQ$ is the diameter of the circle, where $P$ and $Q$ are $(5, 6)$ and $(-2, 2)$ respectively. Find the equation of the circle. \hfill(5 . . .



6 The point $(1, -3)$ lies on the circle $(x-3)^2 + (y+4)^2 = r^2$. Find the value of $r$.

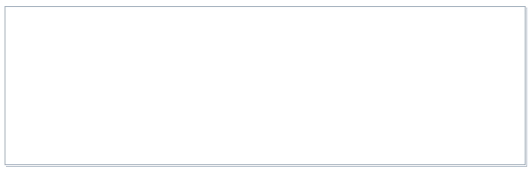


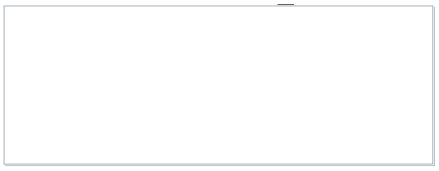
Chapter 6 · Original slide 18

## Exercise 6C - Y1 Page 120

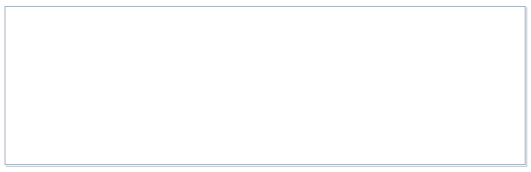
18

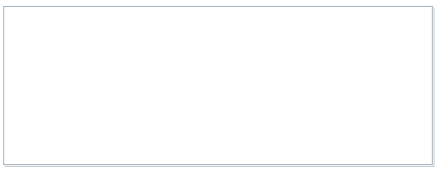
8 a Show that $x^2 + y^2 - 4x - 11 = 0$ can be written in the form $(x-a)^2 + y^2 = r^2$, where $a$ and $r$ are numbers to be found. \hfill(2 marks)
b Hence write down the centre and radius of the circle with equation $x^2 +$
\hfill(2 marks)





9 a Show that $x^2 + y^2 - 10x + 4y - 20 = 0$ can be written in the form $(x-a)^2 + (y-b)^2 = r^2$, where $a, b$ and $r$ are numbers to be found. \hfill(2 m . . .
b Hence write down the centre and radius of the circle with equation $x^2 + y^2 - 10x + 4y - 20 =$
(2 m . . .





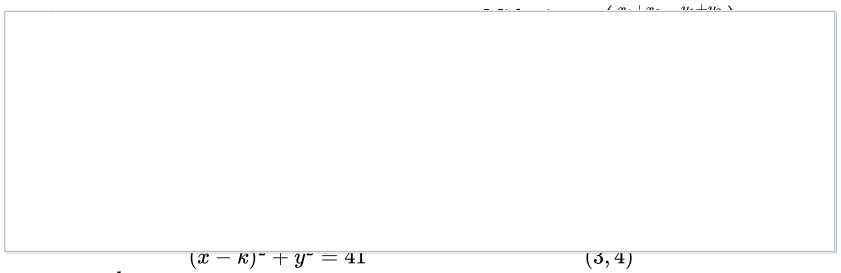
# Coordinate Geometry

Chapter 6 · Original slide 19

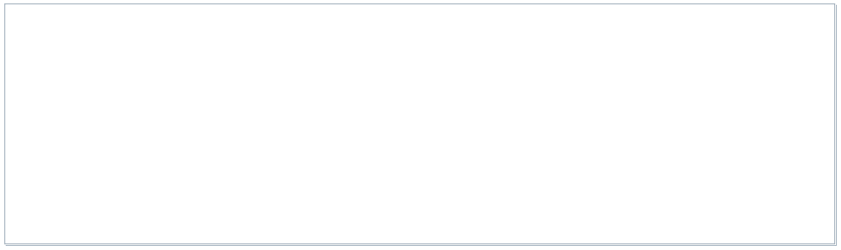
## Exercise 6C - Y1 Page 120

19

12 The point $P(7, -14)$ lies on the circle with equation $x^2 + y^2 + 6x - 14y = 17$. The point $Q$ also lies on the circle such that $PQ$ is a diameter. Find the coordinates of point $Q$.



13 T $(x - k)^2 + y^2 = 41$ $(3, 4)$
possible values of $k$.

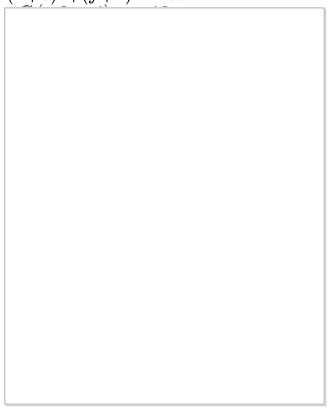


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## Extension

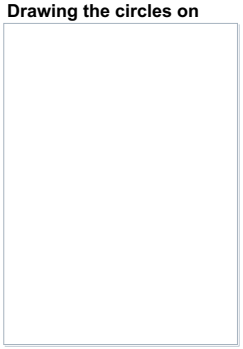
20

1 [MAT 2009 1B] The point on the circle $x^2 + y^2 + 6x + 8y = 75$ which is closest to the origin, is at what distance from the origin?
$(x+3)^2+(y+4)^2=100$



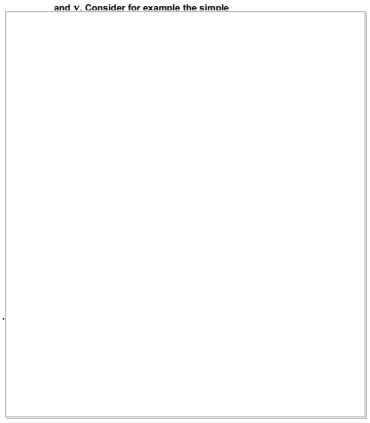
2 [MAT 2007 1D]
The point on the circle $(x-5)^2 + (y-4)^2 = 4$ which is closest to the circle $(x-1)^2 + (y-1)^2 = 1$ has what coordinates?

**Drawing the circles on**



3 [MAT 2016 1I] Let $a$ and $b$ be positive real numbers. If $x^2 + y^2 \leq 1$ then the largest that $ax + by$ can equal is what? Give your expression in terms of $a$ and $b$.

**Many MAT questions consider maximising an expression in terms of x and y. Consider for example the simple**



Coordinate Geometry

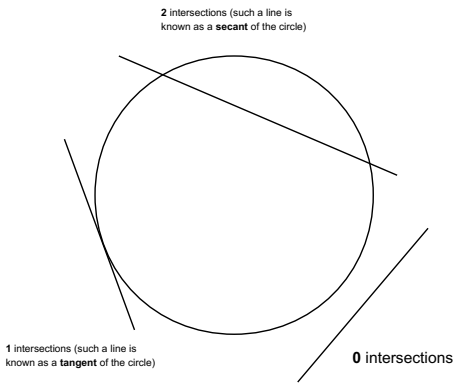
this is the point $(3.4, 2.8)$

Chapter 6 · Original slide 21

## Intersections of Lines and Circles

21

Recall that to consider the **intersection of two lines**, we attempt to solve them **simultaneously** by substitution, potentially using the **discriminant** to show that there are no solutions (and hence no points of intersection).




Show that the line $y = x + 3$ never intersects the circle with equation $x^2 + y^2 = 1$.

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## Test Your Understanding

22

Find the points of intersection where the line $y = x + 6$ meets $x^2 + (y - 3)^2 = 29$.

Using an algebraic (and not geometric) method, determine the $k$ such that the line $y = x + k$ **touches** the circle with equation $x^2 + y^2 = 1$.

# Coordinate Geometry

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## Exercise 6D - Y1 Page 122

23

6 a Show that the line $x + y = 11$ meets the circle with equation $x^2 + (y-3)^2 = 32$ at only one point. \hfill$(4\ \text{mar}\ldots$

b Find the coordinates of the point of intersection. \hfill$(1\ \text{ma}\ldots$

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## Exercise 6D - Y1 Page 122

24

7 The line $y = 2x - 2$ meets the circle $(x-2)^2 + (y-2)^2 = 20$ at $A$ and $B$.

a Find the coordinates of $A$ and $B$.

b Show that $AB$ is a diameter of the circle.

11 The line with equation $y = 4x - 1$ does not intersect the circle with equation $x^2 + 2x + y^2 = k$. Find the range of possible values of $k$.

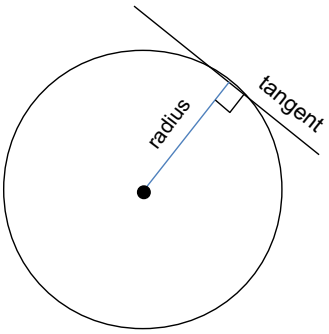
# Coordinate Geometry

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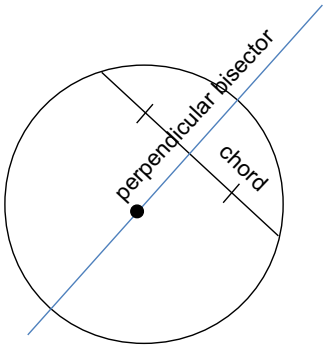
## Tangents, Chords, Perpendicular Bisectors

25

There are two circle theorems that are of particular relevance to problems in this chapter, the latter you might be less familiar with:




The tangent is perpendicular to the radius (at the point of intersection).




The perpendicular bisector of any chord passes through the centre of the circle.

**Why this will help:**
If we knew the centre of the circle and the point of intersection, we can easily find the gradient of the radius, and thus the gradient and hence equation of the tangent.

**Why this will help:**
The first thing we did in this chapter is find the equation of the perpendicular bisector. If we had two chords, and hence found two bisectors, we could find the point of intersection, which would be the centre of the circle.

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## Examples

26

The circle $C$ has equation $(x-3)^2+(y-7)^2=100$.
a) Verify the point $P\,(11,1)$ lies on $C$.
b) Find an equation of the tangent to $C$ at the point $P$, giving your answer in the form $ax+by+c=0$

A circle $C$ has equation $(x-4)^2+(y+4)^2=10$
The line $l$ is a tangent to the circle and has gradient -3. Find two possible equations for $l$, giving your answers in the form $y=mx+c$.

# Coordinate Geometry

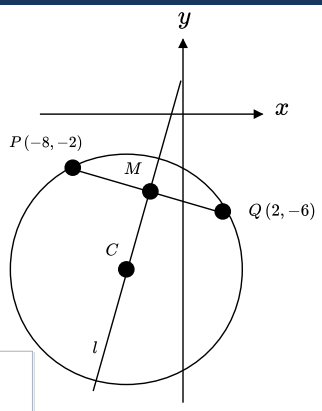
Chapter 6 · Original slide 27

## Determining the Circle Centre

27

The points $P$ and $Q$ lie on a circle with centre $C$, as shown in the diagram. The point $P$ has coordinates $(-8, -2)$ and the point $Q$ has coordinates $(2, -6)$. $M$ is the midpoint of the line segment $PQ$.
The line $l$ passes through the points $M$ and $C$.

a) Find an equation for $l$.
b) Given that the $y$-coordinate of $C$ is -9:
i) show that the $x$-coordinate of $C$ is -5.
ii) find an equation of the circle.




a

b

$-9)$

c

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## Test Your Understanding

28

A circle has centre $C\,(3, 5)$, and goes through the point $P(6, 9)$. Find the equation of the tangent of the circle at the point $P$, giving your equation in the form $ax + by + c = 0$ where $a, b, c$ are integers..

A circle passes through the points $A(0, 0)$ and $B\,(4, 2)$. The centre of the circle has $x$ value -1. Determine the equation of the circle.

# Coordinate Geometry

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## Exercise 6E - Y1 Page 126-128

29

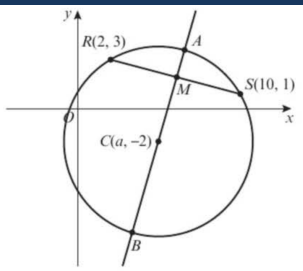
6 The tangent to the circle $(x+4)^2+(y-1)^2=242$ at $(7,-10)$ meets the $y$-axis at $S$ and the $x$-axis at $T$.
a Find the coordinates of $S$ and $T$. \hfill(5 marks)
b Hence, find the area of $\triangle OST$, where $O$ is the origin. \hfill(3 marks)

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## Exercise 6E - Y1 Page 126-128

30

10 The points $R$ and $S$ lie on a circle with centre $C(a,-2)$, as shown in the diagram.
The point $R$ has coordinates $(2,3)$ and the point $S$ has coordinates $(10,1)$.
$M$ is the midpoint of the line segment $RS$.
The line $l$ passes through $M$ and $C$.
a Find an equation for $l$. (4 marks)
b Find the value of $a$. (2 marks)
c Find the equation of the circle. (3 marks)




d Find the points of intersection, $A$ and $B$, of the line $l$ and the circle. (5 marks)

# Coordinate Geometry

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## Exercise 6E - Y1 Page 126-128

31

**Extension:**

1 *[MAT 2012 1A]* Which of the following lines is a tangent to the circle with equation $x^2 + y^2 = 4$?

A) $x + y = 2$
B) $y = x - 2\sqrt{2}$
C) $x = \sqrt{2}$
D) $y = \sqrt{2} - x$
E)



**Mark scheme on next slide.**

2 *[AEA 2006 Q4]* The line with equation $y = mx$ is a tangent to the circle $C_1$ with equation $(x + 4)^2 + (y - 7)^2 = 13$
(a) Show that $m$ satisfies the equation $3m^2 + 56m + 36 = 0$
The tangents from the origin $O$ to $C_1$ touch $C_1$ at the points $A$ and $B$.
(b) Find the coordinates of the points $A$ and $B$.
Another circle $C_2$ has equation $x^2 + y^2 = 13$. The tangents from the point $(4, -7)$ to $C_2$ touch it at the points $P$ and $Q$.
(c) Find the coordinates of either the point $P$ or the point $Q$.
**Mark scheme on next slide.**

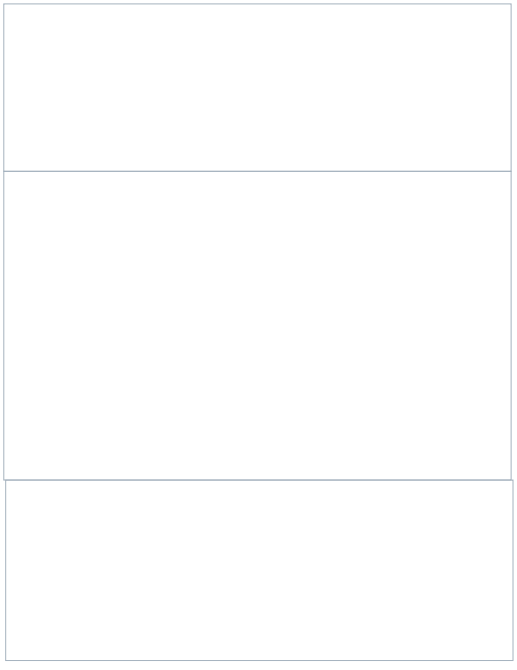
(This is not a tangent/chord question but is worthwhile regardless!)

3 *[STEP 2005 Q6]*
(i) The point $A$ has coordinates $(5, 16)$ and the point $B$ has coordinates $(4, -4)$. The variable $P$ has coordinates $(x, y)$ and moves on a path such that $AP = 2BP$. Show that the Cartesian equation of the path of $P$ is $(x + 7)^2 + y^2 = 100$.
(ii) The point $C$ has coordinates $(a, 0)$ and the point $D$ has coordinates $(b, 0)$. The variable point $Q$ moves on a path such that $QC = k \times QD$, where $k > 1$.
Given that the path of $Q$ is the same as the path of $P$, show that $\frac{a+7}{b+7} = \frac{a^2+51}{b^2+51}$
Show further that $(a + 7)(b + 7) = 100$, in the case $a \neq b$.

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## Mark Scheme for Extension Question 2

32



## Coordinate Geometry

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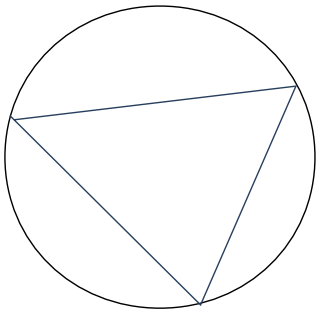
### Mark Scheme for Extension Question 3

33

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### Triangles in Circles

34



We'd say:

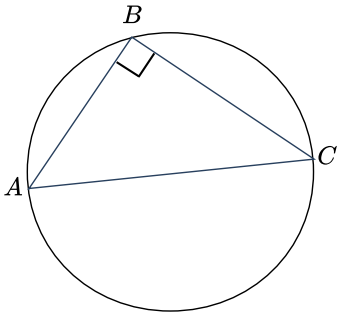
- The triangle **inscribes** the circle.
  (A shape inscribes another if it is inside and its boundaries touch but do not intersect the outer shape)
- The circle **circumscribes** the triangle.
- If the circumscribing shape is a circle, it is known as the **circumcircle** of the triangle.
- The centre of a circumcircle is known as the **circumcentre**.
-

# Coordinate Geometry

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## Triangles in Circles

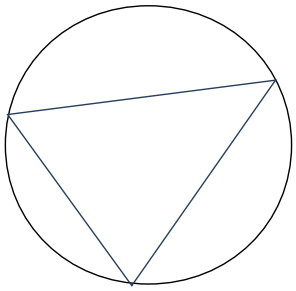
35




If $\angle ABC = 90^\circ$ then:

Similarly if $AC$ is the diameter of a circle:

• $\angle \mathbf{ABC = 90^\circ}$ **therefore** $\mathbf{AB}$



Given three points/a triangle we can find the centre of the circumcircle by:

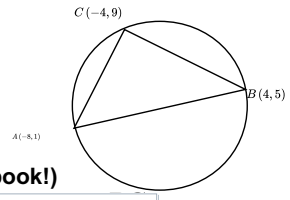
• **Finding the equation of the**

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## Example

36

*[Textbook]* The points $A\,(-8,1), B\,(4,5), C\,(-4,9)$ lie on a circle.
a) Show that $AB$ is a diameter of the circle.




**Method 1:**

**Method 2: (not in textbook!)**

b) Hence find the equation of the circle.

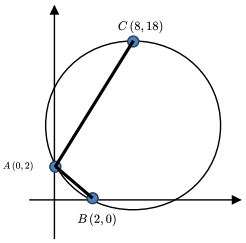
# Coordinate Geometry

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## Example

37

The points $A\,(0,2), B\,(2,0), C\,(8,18)$ lie on the circumference of a circle. Determine the equation of the circle.

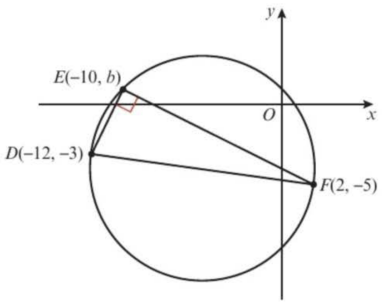



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## Exercise 6F - Y1 Page 131-132

38

8 The points $D(-12,-3), E(-10,b)$ and $F(2,-5)$ lie on the circle $C$ as shown in the diagram.
Given that $\angle DEF = 90^\circ$ and $b > 0$
a show that $b = 1$ \hfill(5 marks)
b find an equation for $C$. \hfill(4 marks)



# Coordinate Geometry

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## Exercise 6F - Y1 Page 131-132

39

9 A circle has equation $x^2 + 2x + y^2 - 24y - 24 = 0$.
a Find the centre and radius of the circle.
b The points $A(-13, 17)$ and $B(11, 7)$ both lie on the circumference of the circle. Show that $AB$ is a diameter of the circle.
c The point $C$ lies on the negative $x$-axis and the angle $ACB = 90°$. Find the coordinates of $C$.

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## Exercise 6F - Y1 Page 131-132

40

**Extension:**

1 *[STEP 2009 Q8 Edited]* If equation of the circle $C$ is $(x - 2t)^2 + (y - t)^2 = t^2$, where $t$ is a positive number, it can be shown that $C$ touches the line $y = 0$ as well as the line $3y = 4x$.

Find the equation of the incircle of the triangle formed by the lines $y = 0$, $3y = 4x$ and $4y + 3x = 15$.
**Note**: The incircle of a triangle is the circle, lying totally inside the triangle, that touches all three sides.

**Solution:** $\mathbf{(x-2)^2+(y-1)^2=1}$